

LESSON 5-4 • SESSION 2 • PRACTICE • FORM A

Composite Figures

Learning target – I can find the area of a composite figure by adding or subtracting the areas of simpler shapes. · 6.G.A.1

REMEMBER

- A **composite figure** is made of simpler shapes: rectangles, triangles, squares.
- **Add** the areas of pieces that are joined. **Subtract** the area of a piece that is cut out.
- Use each piece's own formula: rectangle $l \times w$, triangle $\frac{1}{2} \times b \times h$.

WORKED EXAMPLE

A 14 cm by 9 cm rectangle has a triangle (base 6 cm, height 4 cm) cut out.

Step 1 Whole rectangle: $14 \times 9 = 126 \text{ cm}^2$.

Step 2 Cut-out triangle: $\frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$.

Step 3 Removed, so subtract: $126 - 12$.

Answer **114 cm² left**

WORDS **composite figure** (*figura compuesta*) · **add** (*sumar*) · **subtract** (*restar*)

PRACTICE

show your work · label units

1. Decide whether to add or subtract. Then find each area.

a. 40 cm² rectangle + 12 cm² triangle ____ cm² b. 100 in² square, 16 in² cut out ____ in²

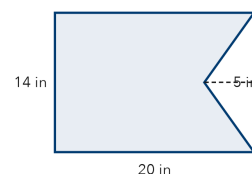
c. 9 ft square, 3 ft square cut out ____ ft² d. 10 m × 4 m, 2 m square cut out ____ m²

2. **Swallowtail Flag.** A flag starts as a rectangle 20 in by 14 in. A triangle notch 5 in deep is cut out of the right edge.

a. Area of the whole rectangle ____ in²

b. Area of the notch ____ in²

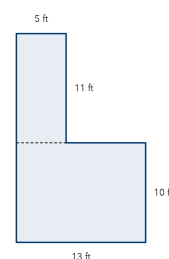
c. Area of the flag ____ in²



3. **L-Shaped Room.** An L-shaped floor splits into a 13 ft by 10 ft rectangle and a 5 ft by 11 ft rectangle. Find the floor area.

Show your work.

Answer _____ ft²



4. **Wall and Window.** A wall is 22 ft long and 10 ft tall. A square window 5 ft on each side is not painted. Find the painted area.

Hint The window is taken away.

Show your work.

Answer _____ ft²

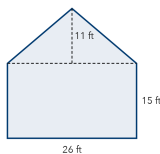
5. **Plywood Sheet.** A plywood sheet is 24 ft by 9 ft. A 6 ft by 3 ft opening is cut out. How much plywood is left?

Show your work.

Answer _____ ft²

6. **House Front.** A house front is a rectangle with a triangle roof. Find the total area.

Hint Use each shape's own formula, then add.



Show your work.

Answer _____ ft²

7. **Deja's Poster.** A poster is 32 in by 25 in. A 10 in square is cut from one corner. Deja says 800 in² is left. Is Deja right?

Hint Did the corner square get taken away?

Show your work.

Area left _____

8. **Playground Plan.** A playground is 30 m by 20 m. Rubber covers all of it except a triangle sandbox (base 12 m, height 10 m). Rubber costs \$14 per m².

- a. Area of the whole rectangle _____ m²
- b. Area covered with rubber _____ m²
- c. Cost of the rubber _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Omar's Work.** A wall is 19 ft by 11 ft with a 5 ft by 5 ft vent that is not painted. Omar wrote: "Painted area = 209 + 25 = 234 ft²." Explain the mistake and fix it.

TEST PRACTICE

choose the best answer

10. When a piece is cut out of a figure, you –
- A** add its area
 - B** subtract its area
 - C** multiply by its area
 - D** ignore it
11. A composite figure is made of –
- A** only triangles
 - B** one rectangle
 - C** two or more simpler shapes
 - D** curved lines only

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

700 A	7,560 T	540 C	315 Q	185 U	198 T	245 S	900 Z	533 R	195 B
#2c	#3	#4	#5	#6	#7	#8b	#8c		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____